Check for updates

Research Article

Psychological Science
24(8) 1446–1455
© The Author(s) 2013
Reprints and permissions:
sagepub.com/journalsPermissions.nav
DOI: 10.1177/0956797612472203
pss.sagepub.com

Predictive and Reactive Mechanisms in Smile Reciprocity

Erin A. Heerey and Helen M. Crossley
Bangor University

Abstract

During face-to-face interactions, people reciprocate their conversation partners' genuine and polite smiles with matching smiles. In the research reported here, we demonstrated that predictive mechanisms play a role in this behavior. In natural interactions (Study 1), participants anticipated a substantial proportion of genuine smiles but almost no polite ones. We propose that reinforcement-learning mechanisms underpin this social prediction and that smile-reciprocity differences arise because genuine smiles are more rewarding than polite smiles. In Study 2, we tested this idea using a learning task in which correct responses were rewarded with genuine or polite smiles. We measured participants' smile reactions with electromyography (EMG). As in natural interactions, people mimicked polite smiles reactively, after seeing them appear. Interestingly, the EMG data showed predictive responding to genuine smiles only. These results demonstrate that anticipating social rewards drives predictive social responding and therefore represent a significant advance in understanding the mechanisms that underpin the neural control of real-world social behavior.

Keywords

social interaction, reciprocity, smiles, prediction, reward learning, rewards, electrophysiology

Received 2/12/12; Revision accepted 11/30/12

Social interactions involve highly coordinated exchanges of verbal and nonverbal information. These exchanges are often reciprocal, meaning that interaction partners trade behavior in a like-for-like fashion, responding positively to positive cues and negatively to negative ones (Cialdini & Goldstein, 2004; King-Casas et al., 2005). Smiles, for example, are frequently reciprocated social actions (Capella, 1997; Heerey & Kring, 2007; Hess & Bourgeois, 2010). Research has shown that in natural interactions, people return their partners' smiles with high probability, responding to the majority of their partners' smiles with smiles (Capella, 1997; Hess & Bourgeois, 2010; Wild, Erb, Eyb, Bartels, & Grodd, 2003). Indeed, failing to reciprocate smiles reduces partner-reported positive affect and interaction quality (Capella, 1997; Heerey & Kring, 2007), which suggests that people expect their interaction partners to return smiles and find it aversive when they do not.

Broadly speaking, there are two major classes of smile. Genuine smiles of pleasure occur spontaneously during episodes of positive affect and involve the action of the orbicularis oculi muscle (Ekman, Davidson, & Friesen, 1990). Polite smiles, which rarely involve the orbicularis oculi (Ekman et al., 1990; but see Krumhuber & Manstead, 2009) are not pleasure related and serve, for example, as tokens of politeness when sociocultural norms require a smile (Ekman, Sorenson, & Friesen, 1969; Goffman, 1955). People use both types of smiles frequently in naturalistic interactions (Hess & Bourgeois, 2010), but what is more interesting is that people also reciprocate both types of smiles with high fidelity. We previously found that in face-to-face interactions, people's accuracy in matching their conversation partners' smiles with smiles of the same type is above 90% (Heerey & Kring, 2007).

Some research has suggested that smile reciprocity is a form of social mimicry (Achaibou, Pourtois, Schwartz, & Vuilleumier, 2008; Hess & Bourgeois, 2010)—in other words, that people reciprocate smiles reactively, after seeing them produced. However, because genuine smiles

Corresponding Author:
Erin A. Heerey, School of Psychology, Brigantia Building, Bangor University, Penrallt Road, Bangor, Gwynedd LL57 2AS, Wales
E-mail: e.heerey@bangor.ac.uk

have a higher social-reward value than do polite ones (Shore & Heerey, 2011), we asked whether genuine-smile reciprocity might also be predictive, as responses to nonsocial rewards are. For example, in reinforcement-learning models, which research has suggested may be important in the on-line control of social behavior (Behrens, Hunt, & Rushworth, 2009; King-Casas et al., 2005), individuals learn to anticipate forthcoming rewards (Schultz, 2007). If genuine smiles are indeed a form of social reward, people should be more likely to anticipate genuine smiles than the relatively less rewarding polite ones.

To learn whether there are differences in the degree to which people predict genuine and polite smiles, we examined *smile onset asynchronies* (the time lag between the onset of an individual's smile and the onset of his or her conversation partner's smile) in naturalistic, face-to-face interactions. If smile reciprocity is purely reactive (e.g., if it is a form of mimicry rather than reciprocity; Chartrand & Bargh, 1999; Stel & van Knippenberg, 2008), then the onset asynchronies for most returned smiles should be longer than 200 ms, the minimum time required to process and respond to a complex stimulus with a complex voluntary movement (Sanders, 1998) such as a smile. However, the presence of a substantial proportion of reciprocated smiles with onset times faster than 200 ms would provide evidence for predictive social responding.

We probed smile reciprocity both in face-to-face interactions (Study 1) and under laboratory conditions (Study 2). Although laboratory studies are more controlled than naturalistic interactions, the constraints of the experimental context reduce their ecological validity. Social stimuli are particularly vulnerable to this problem (Repacholi & Slaughter, 2003), which means that conclusions based on laboratory data alone may generalize poorly to real-world conditions (Mitchell, 2012). The inclusion of data from naturalistic interactions thus helps to bridge the gap between natural social behavior and its underlying mechanisms.

Study 1

To determine whether smile reciprocity in natural interactions is reactive mimicry or predictive reciprocity, we examined smile onset asynchronies using data from forty-eight 5-min dyadic social interactions between strangers of the same sex ($N = 96$; half of the dyads were composed of male participants and the other half were composed of female participants). Dyad members were simply instructed to spend 5 min getting to know one another and were told that they could discuss whatever they wished. Interactions were videotaped for later analysis.

Coding

Videos were coded for the presence of genuine and polite smiles using a nonverbal coding system based on the Facial Expression Coding System (Kring & Sloan, 2007). Briefly, in this system, smiles are considered genuine when both the zygomaticus major and the orbicularis oculi are active, and polite when they involve only the zygomaticus (Ekman et al., 1990). Four coders who were blind to the study's hypotheses and trained to a high standard (> 95% correct on a set of training videos) independently coded the video data. To avoid bias, we had coders view videos from one participant at a time and code the entire 5-min interaction before moving on to another participant. Partners' videos were later linked using time stamps. Coders classified each smile they saw as either genuine or polite and recorded the time stamp of the first and the last frame at which the smile was visible (the smile's onset and offset, respectively). Videos were stripped of their audio content to ensure that speech content did not affect nonverbal-behavior coding.

To assess interrater agreement, we had two independent coders code a pseudorandom sample of 40 of the 96 videos, such that each coder's set of videos overlapped with that of each other coder and 5 videos were coded by all four coders. Cohen's kappa coefficients, used to assess agreement for both smile type and smile onset, served as our measure of interrater reliability (see Bakeman & Quera, 2011). Disagreements about onset and offset times were resolved by consensus. Before consensus coding, coders achieved excellent agreement for both smile types (genuine smiles: 93%; polite smiles: 89%).

Data analysis

Videos were recorded at a rate of 25 frames per second, so time estimates had a 40-ms resolution. To determine smile onset asynchrony, we calculated the time elapsed between the onset of a conversation partner's smile and the participant's response. To increase the probability that participants' response smiles were related to their partners' initiating smiles, rather than to other events, we included only exchanges in which participants reciprocated their partners' smiles within 4 s.

Across the 48 interactions, we recorded 619 smile exchanges. We excluded 47 exchanges (7.5%) in which participants incorrectly returned the initiating smile, because nonreciprocity is a different type of social signal than a returned smile. This left 572 smile exchanges, 56.58% of which involved genuine smiles. We then determined the number of smiles of each type that occurred more and less than 200 ms after the onset of the conversation partner's smile.

Results

Figure 1 shows the distribution of smile onset asynchronies for genuine and polite smiles. There were substantial differences in the frequency of anticipatory reciprocity, depending on whether the smile exchange involved genuine or polite smiles: 21% of genuine smiles ($SD = 0.21\%$) and 7% of polite smiles ($SD = 0.15\%$) had onset asynchronies faster than 200 ms. Genuine smiles were statistically more likely to be anticipated than were polite smiles, paired-samples $t(92) = 5.58$, $p < .001$.[1] The median onset asynchrony for exchanged genuine smiles was 780 ms (SD = 74 ms). Participants responded to polite smiles more slowly (median onset asynchrony: 1,170 ms, SD = 93 ms). A paired-samples t test confirmed that people reciprocated genuine smiles more quickly than polite smiles, $t(92) = 3.89$, $p < .001$. Thus, results suggested that participants were more likely to anticipate genuine smiles and to reactively mimic polite ones.

Discussion

These results demonstrate that in face-to-face social environments, individuals both reactively mimic and anticipate their partners' expressions. More interestingly, these data hint that in naturalistic interactions, anticipatory reciprocity is substantially more likely in response to genuine smiles than polite smiles.

One explanation for this finding might relate to extraneous variables, such as speech content or differences in smile predictability during interactions. For example, when becoming acquainted, people may share anecdotes or jokes (Eder, 1993). Such content, which is often associated with genuine smiles (Fridlund, 1991; Russell, Bachorowski, & Fernandez-Dols, 2003), although it also relates to polite smiles (Eder, 1993), may be highly predictable in the timing of its delivery (Pickering et al., 2009) and may thereby allow individuals to predict some genuine smiles.

Additionally, differences in the anticipation of polite and genuine smiles might arise because of differences in their social values (Averbeck & Duchaine, 2009; O'Doherty et al., 2003; Scharlemann, Eckel, Kacelnik, & Wilson, 2001). For example, we have previously shown that participants value genuine smiles to the extent that they are willing to forgo the chance to win money in order to see them (Shore & Heerey, 2011). Differences in participants' preferences for these smiles may in turn underpin differences in anticipatory activity when participants expect preferred-smile feedback (Schultz, 2006; Watanabe, 1996). Thus, participants should behaviorally anticipate predicted genuine smiles but not predicted polite ones. To test this hypothesis, we examined smile reciprocity under laboratory conditions using a sensitive measure of facial behavior: electromyography (EMG).

Study 2

We examined responses to genuine and polite smiles in the context of a probabilistic learning task with social stimuli. Importantly, this laboratory task ensured that genuine and polite smiles were equally predictable and unaffected by extraneous variables that play a role in natural smile reciprocity.

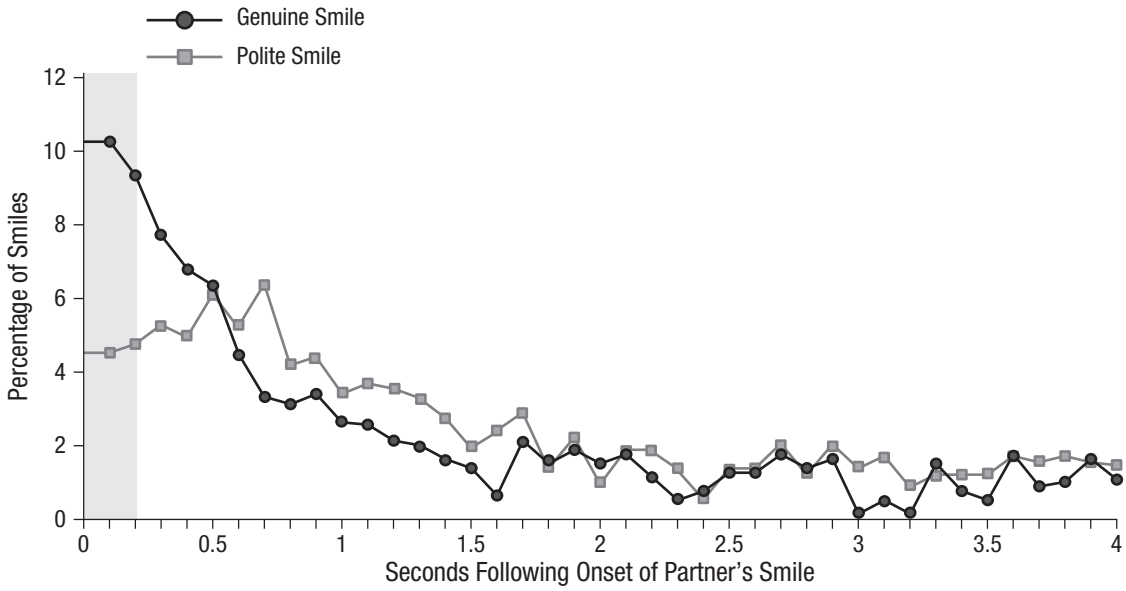

Fig. 1. Results of Study 1: distributions of smile onset asynchronies in face-to-face interactions as a function of smile type. The shaded region shows onset asynchronies of 200 ms or less.

Method

Participants. Thirty-five right-handed psychology undergraduates (85% female, 15% male; mean age = 21.68 years, *SD* = 1.12) participated in return for partial course credit. There were no financial incentives for participating. All participants provided written informed consent, and Bangor University's ethics committee approved the study.

Procedure. Skin surfaces were cleansed and then scrubbed with electrode gel and a mildly abrasive pad to reduce impedance. We then placed a single reference electrode in the center of participants' foreheads near the hairline and placed pairs of recording electrodes (1-cm interelectrode distance), oriented parallel to the muscle-fiber bodies, over three muscle sites (corrugator supercilii, orbicularis oculi, zygomaticus major) on the left side of participants' faces using the anatomical guidelines suggested in Fridlund and Cacioppo (1986).

We measured electrical impedance after giving the electrodes 10 min to adjust. Any electrode pair with a resistance of > 10 kΩ was removed and replaced following repetition of the skin-preparation procedure. We excluded data from 1 male participant with suboptimal resistance values (> 46 kΩ after electrode replacement).

Using a computer, participants completed a simple probabilistic-learning task in which they learned, via trial and error, to associate each of four faces with a key press. On each trial, participants saw a fixation cross for 2,000 ms, followed by the presentation of a neutral face. Using a set of two keys on the keyboard (the "z" key and the "m" key), they made either a left- or right-key press during the presentation of the neutral face. Once the computer registered the key press, a frame appeared around the face they had selected to indicate that feedback would be forthcoming. If the correct button had been pressed, two of the faces smiled genuinely and two smiled politely. If a response was incorrect, the faces frowned.

To give participants the opportunity to anticipate expressive displays, we implemented a 4,000-ms delay between the end of the response window and the onset of the expression. The neutral face was continuously present during this delay. After the delay, the computer displayed six frames (presented for 50 ms each) showing the face transitioning from a neutral expression to the peak expression. The peak expression was visible for 4,000 ms, which allowed us to measure reactive facial mimicry. A blank screen (4,000 ms) indicated the end of the trial. The average reaction time across all trials was 1,092 ms (*SD* = 260 ms, range = 498–2,659 ms).

We chose a probabilistic task because it allowed us to measure participants' responses to smiles that differed in the degree to which they were predictable. To maximize the chance of finding anticipatory smiles of both types, we included a deterministic condition in which all smiles were perfectly predictable. Two faces (one genuinely smiling and one politely smiling) displayed smiles on 100% of correct responses (one in response to presses of the left key, one in response to presses of the right key). Because people reciprocate approximately 70% of polite smiles (Heerey & Kring, 2007), we had the remaining faces (one genuinely smiling and one politely smiling) display smiles on 70% of correct responses (one in response to presses of the left key and one in response to presses of the right key), so that on 30% of the trials on which participants made the correct response, they saw a frown instead of a smile. We used this condition to ensure that genuine-smile reciprocity was not simply related to expectations about smile-return frequency. Half the participants viewed male faces during the task, and the others viewed female faces (counterbalanced across participant gender). We counterbalanced face identity across probability, smile-type, and key-press conditions. An independent sample of 48 participants had previously rated all the faces as similarly attractive (for a full description of stimulus characteristics, see Shore & Heerey, 2011).[2]

We instructed participants to earn as many smiles as possible during the task by learning which response was most closely associated with each face. Participants completed 120 trials divided into three task blocks. There were 10 presentations of each face per block, in random order. The task was programmed in E-Prime (Psychology Software Tools, Sharpsburg, PA).

Electromyography. EMG signals were recorded continuously throughout the task and acquired using MP36 hardware and AcqKnowledge Software (Biopac, Goleta, CA). Signals were detected using a common-mode-rejection algorithm, which subtracts signals at the reference site from those at the recording sites to remove common noise. Electrodes were 4-mm shielded Ag/AgCl surface electrodes filled with conductive gel. We sampled at a rate of 2,000 Hz with a gain of 5,000 Hz and applied a 50-Hz notch filter to minimize electrical noise.

Signal processing. We processed the EMG data in several steps. First, the data were bandpass filtered using a fourth-order Butterworth filter (10–400 Hz). We use 400 Hz as our low-pass cutoff because frequencies greater than 400 Hz contribute negligibly to facial EMG signals (van Boxtel, 2003). Data were then full-wave rectified and log-transformed to reduce the influence of extreme values. Finally, to allow comparisons across participants and muscle sites, we standardized each participant's data to the activity at fixation (baseline) within muscle sites. All data processing and analysis utilized

purpose-written functions in MATLAB (The Mathworks, Natick, MA).

Data analysis. To estimate the point in the task at which participants had learned the correct stimulus-response mappings, we determined when participants had achieved three consecutive correct responses to a stimulus. This *trials-to-criterion* measure served as the measure of learning in our behavioral analyses. We calculated separate criteria for each stimulus and subjected the trials-to-criterion data to a repeated-measures analysis of variance (ANOVA) to determine whether smile type (genuine, polite) and reward probability (100%, 70%) influenced learning rates.

To analyze the EMG data, we calculated the average activity for each muscle at each trial phase (neutral face, anticipation, feedback) for each expression type. Because there were only eight instances of incorrect responses or frowns in the data set for deterministic faces, we analyzed only genuine and polite smiles in the 100% condition. In the 70% condition, we analyzed all three expression types (genuine smile, polite smile, frown). We used a repeated measures ANOVA to examine recording site (corrugator supercilii, orbicularis oculi, zygomaticus major), expression type, and trial phase for trials after participants had learned a stimulus to criterion. All post hoc analyses used Bonferroni's Type I error correction.

Results

Behavioral results. We hypothesized that the observed differences in face-to-face smile reciprocity may have been caused by differences in the social values of polite and genuine smiles (Shore & Heerey, 2011). As a corollary, we expected participants to learn the more valuable, genuinely smiling faces more quickly than less valuable, politely smiling ones. Indeed, we observed a main effect of smile type, $F(1, 33) = 4.93$, $p = .03$, $\eta_p^2 = .13$ (Fig. 2a), which showed that participants learned genuinely smiling faces faster than they learned politely smiling faces. A trial-by-trial analysis of the 100% stimuli (Fig. 2b) showed that although participants reached similar levels of performance for genuine and polite smiles, they took longer to do so for polite smiles, $F(1, 33) = 4.62$, $p = .04$, $\eta_p^2 = .12$. Unsurprisingly, we also found that participants learned the stimulus-response mappings to 100% stimuli more quickly than they learned the stimulus-response mappings to 70% stimuli, $F(1, 33) = 7.85$, $p = .008$, $\eta_p^2 = .19$. There was no Probability × Smile Type interaction, $F(1, 33) = 1.87$, $p = .18$, $\eta_p^2 = .05$. Thus, these data suggest

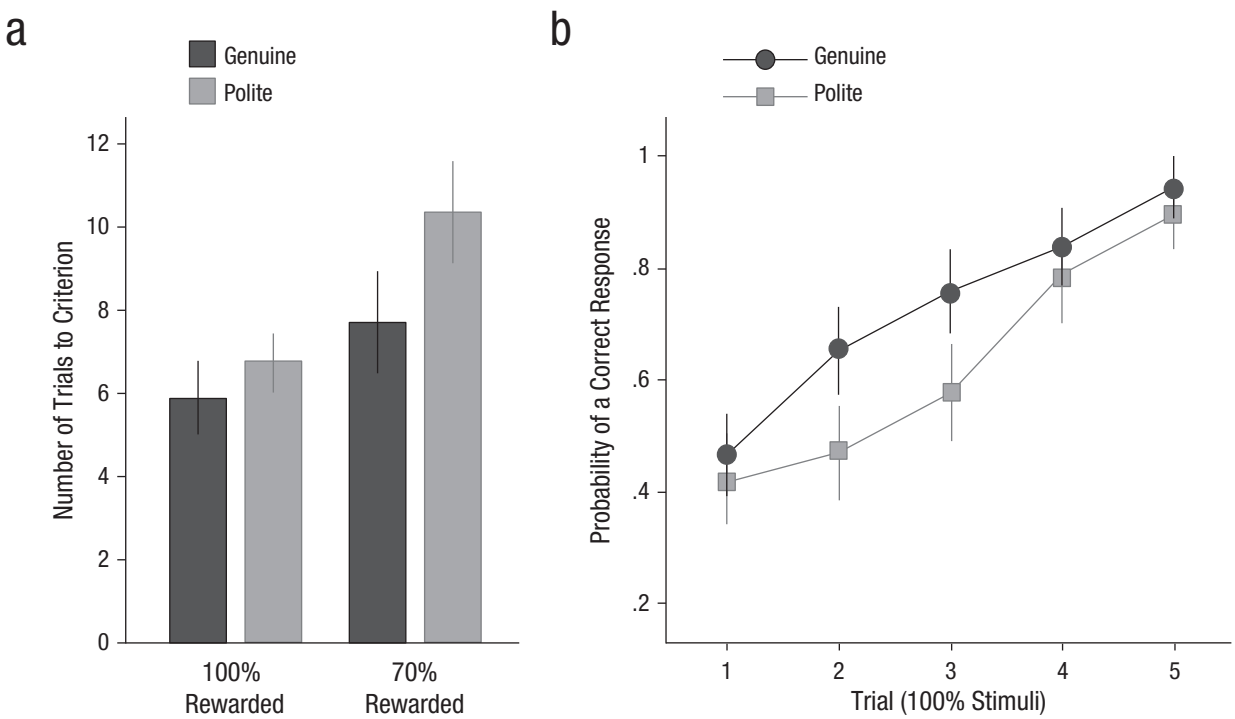

Fig. 2. Behavioral results from Study 2. The graph in (a) shows the number of trials participants completed before reaching stable performance (three consecutive correct responses to a given stimulus) as a function of the probability of receiving a smile after a correct response and smile type. The graph in (b) shows the probability of a correct response over the first five consecutive trials for 100% stimuli (faces that displayed genuine or polite smiles after correct responses with 100% probability) as a function of trial number and smile type. Error bars show ±1 *SEM*.

an important difference in the degree to which genuine and polite smiles guide learning. One explanation for this finding is that the greater social value of genuine smiles relative to polite smiles enhances learning.

EMG results. An omnibus ANOVA examining EMG activity across probability conditions (100%, 70%), muscles (corrugator supercilii, orbicularis oculi, zygomaticus major), trial phases (neutral face, anticipation, feedback), and smile types (genuine, polite; note that in order to examine the main effects of probability condition, we excluded frowns from this analysis) showed that EMG activity was significantly stronger in the 100% condition than in the 70% condition, $F(1, 30) = 7.03$, $p = .01$, $\eta_p^2 = .19$. However, probability condition did not interact with any other factors (ps > .38).

We therefore analyzed the 100% and 70% conditions independently in order to include frown outcomes in the 70% condition. We predicted that participants would specifically mimic visible smiles during the feedback phase. As Figures 3 and 4 show, participants did indeed mimic smiles. As predicted, during smile feedback for both 100% and 70% stimuli, we found significant Expression Type × Muscle interactions—100% stimuli: $F(2, 62) = 9.31$, $p = .005$, $\eta_p^2 = .23$; 70% stimuli: $F(4, 120) = 7.47$, $p < .001$, $\eta_p^2 = .20$. Participants produced similar zygomaticus activity for both smiles (ps > .43) but greater orbicularis oculi activity during genuine smiles compared with polite smiles (ps < .02). Additionally, in the 70% condition, participants generated greater zygomaticus activity when viewing both types of smiles relative to frowns (ps < .02) and more corrugator activity when viewing frowns than both types of smiles (ps < .01). These results confirm previous findings showing that people reactively mimic the expressive displays they see (Chartrand & Bargh, 1999; Stel & Vonk, 2010) and that they reciprocate particular types of smiles specifically.

On the basis of our prediction that people would behaviorally anticipate genuine smiles, we expected elevated zygomaticus and orbicularis oculi activity when participants anticipated genuine smiles (while viewing neutral displays) after learning the task. In contrast, we predicted no such activity during polite-smile anticipation. To test this idea, we examined muscle activity

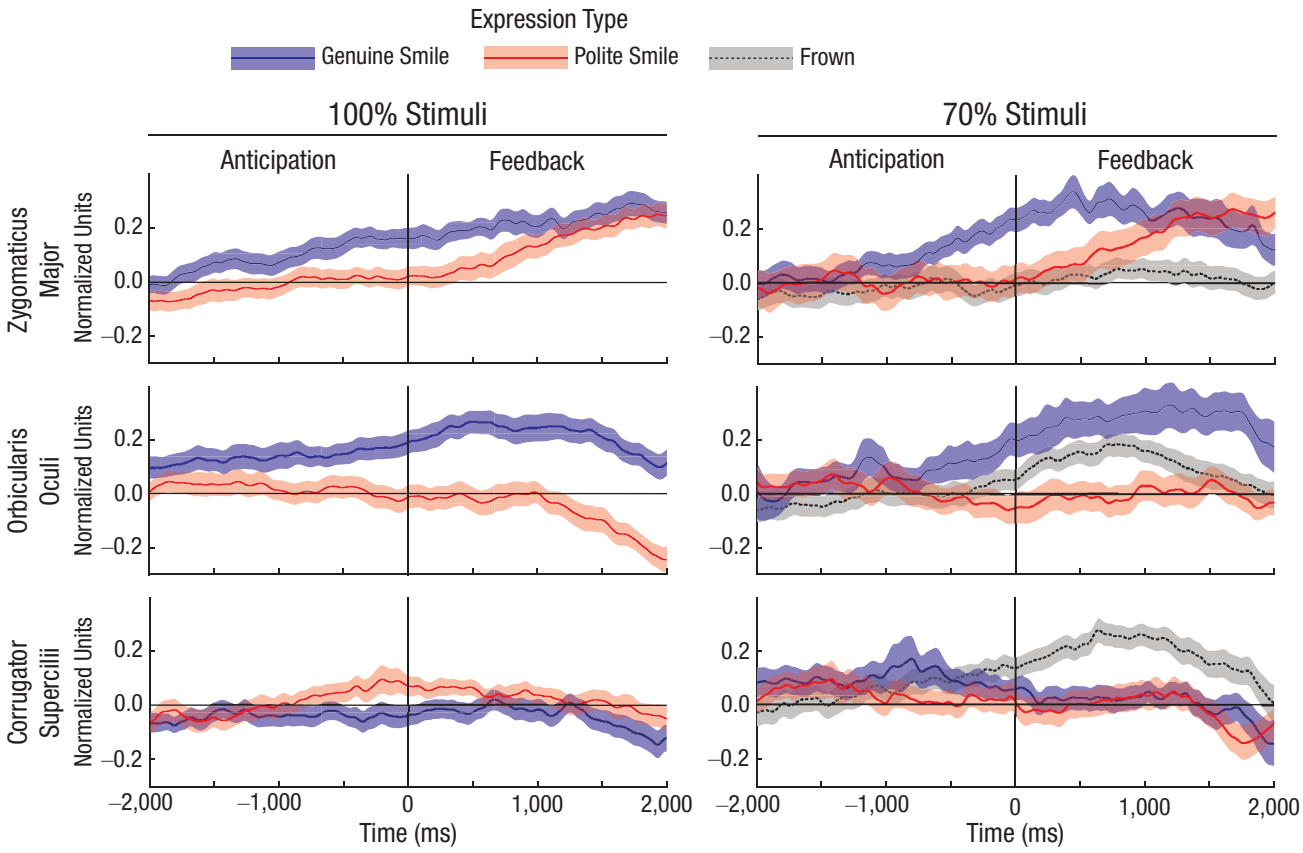

Fig. 3. Results from Study 2: electromyographic (EMG) activity over time. The graphs show changes in EMG traces in standardized units across the smile-anticipation and smile-feedback trial phases as a function of expression type viewed, shown separately for conditions in which face stimuli displayed smiles after correct responses with 100% or 70% probability. (For ease of presentation, traces have been clipped to a 4-s window; full EMG results are presented in Fig. 4). Shaded regions show ±1 *SEM*.

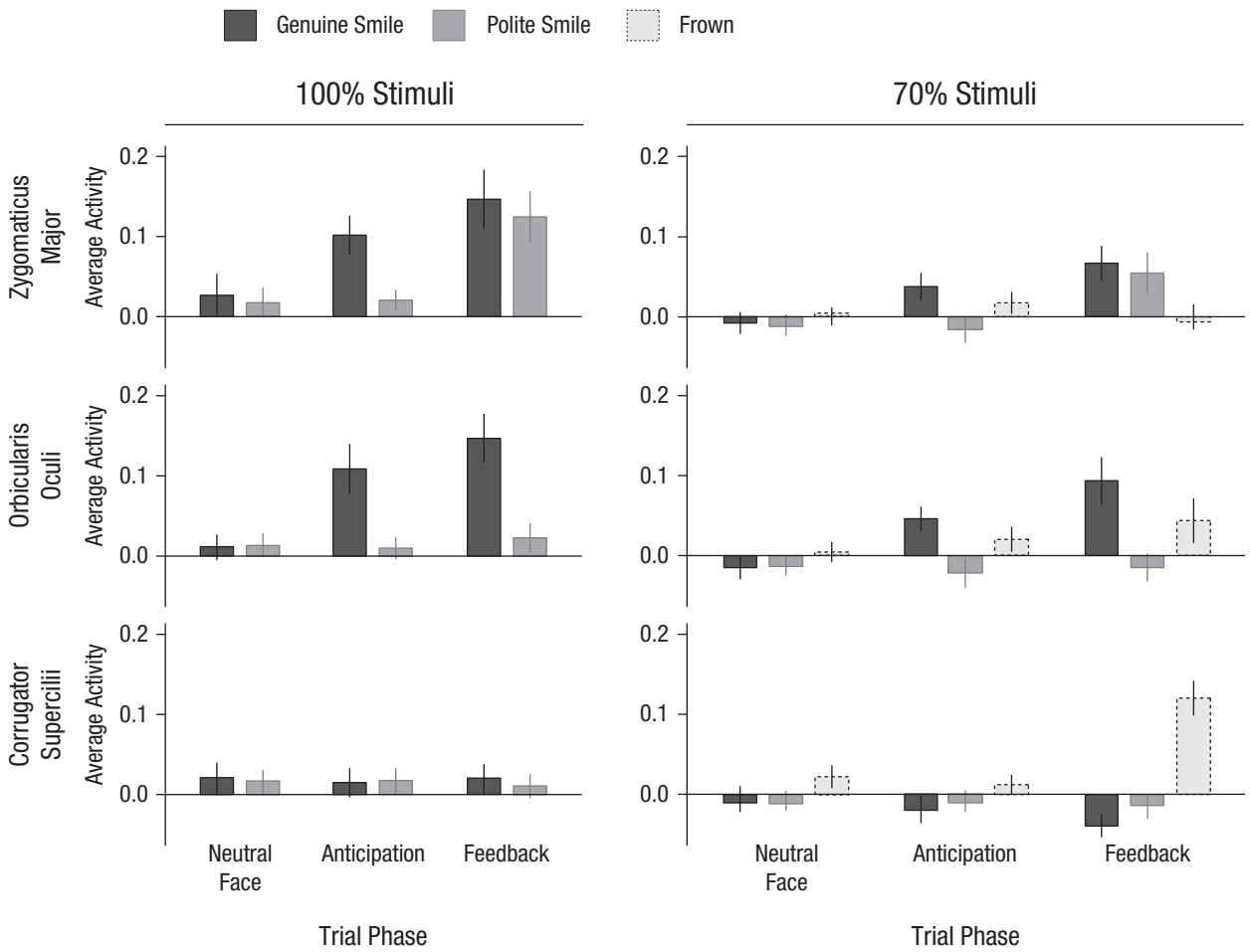

Fig. 4. Electromyographic (EMG) results from Study 2: average activity for the zygomaticus major, orbicularis oculi, and corrugator supercilli as a function of trial phase and expression type, shown separately for conditions in which face stimuli displayed smiles after correct responses with 100% or 70% probability. Error bars show ±1 *SEM*.

during the anticipatory trial phase, while participants awaited feedback.

As predicted, results showed significant Muscle × Expression Type interactions—100% stimuli: $F(2, 62) = 8.41$, $p = .007$, $\eta_p^2 = .21$; 70% stimuli: $F(4, 120) = 2.56$, $p = .04$, $\eta_p^2 = .08$ (see Fig. 3). For the 100% stimuli, participants showed orbicularis oculi and zygomaticus activity greater than baseline level (ps < .007) when they expected genuine smiles. There was no anticipatory activity for polite smiles (ps > .22). Orbicularis oculi activity and zygomaticus activity were also greater for genuine relative to polite smiles during smile anticipation (ps < .02). We found similar results in the 70% condition: Both the zygomaticus and the orbicularis oculi were activated relative to baseline when participants anticipated genuine smiles (ps < .02). This was not true when participants anticipated polite smiles (ps > .63). Additionally, participants showed significant anticipatory activity at the orbicularis oculi recording site when anticipating genuine as opposed to polite smiles ($p = .03$) and tended to do so at the zygomaticus site as well ($p = .06$). Differences in corrugator activity across expression types were not significant (ps > .21). Thus, these data provide clear evidence of anticipatory activity for genuine, but not polite, smiles.

Discussion

In addition to replicating previous research showing reactive mimicry of observed facial emotion (Heyes, 2011; van Baaren, Holland, Kawakami, & van Knippenberg, 2004), Study 2 yielded two important results. First, participants learned stimulus-response mappings more quickly when they were reinforced with genuine, compared with polite, smiles. Second, like our naturalistic data, our laboratory data showed that participants made anticipatory responses when expecting genuine smiles but not polite smiles. Although differences in the social-reward values of the two types of smiles can directly explain learning-rate

differences, reinforcement-learning processes cannot directly explain anticipatory genuine smiling, given that at the time of EMG recording, participants could perfectly predict which smile they would see. To explain anticipatory genuine-smile reciprocity, we suggest that participants' preference for genuine smiles led to smile-specific anticipatory activity.

To confirm that differences in anticipatory EMG activity were related to social reinforcement specifically, rather than to reward expectation more generally, we conducted a second EMG experiment that included genuine and polite smiles in addition to monetary rewards. This allowed us to determine whether anticipatory activity occurred in response to monetary rewards as well as social ones. Anticipatory EMG activity occurred only when participants expected genuine smiles, rather than generalizing to monetary reinforcement (for more details, see the Supplemental Material available online). Our results therefore suggest that the social value of genuine smiles, rather than reward value generally, drives anticipatory responding.

General Discussion

Together, the present results suggest that both domain-general and domain-specific reward-learning mechanisms support humans' ability to achieve the finely coordinated social behavior they produce. Our behavioral data showed that participants learned correct stimulus-response mappings faster for genuinely smiling faces than for politely smiling faces. Domain-general reinforcement-learning mechanisms easily explain this difference, given that participants learned the more rewarding, genuine smiles faster than the less rewarding, polite smiles. However, the EMG results showed clear evidence of domain-specific anticipatory responding to the highly socially valuable genuinely smiling faces, a finding consistent with research showing that reward preferences drive differences in anticipatory neural activity (Schultz, 2007; Watanabe, 1996). This was true both when smiles were 100% predictable and when they were predictable only 70% of the time.

It is noteworthy that, despite significant differences between the naturalistic and laboratory designs, our data from face-to-face interactions showed that anticipatory responding was stronger for genuine smiles than for polite smiles in both settings. Thus, the agreement between experimental and observational findings suggests that anticipatory genuine-smile reciprocity is an important element of real-world social behavior. The results of Study 1 show that natural-smile reciprocation is fast and precise. Pairs of participants specifically returned genuine and polite smiles, even when given no instruction about how to behave. The speed with which they returned genuine smiles constitutes clear evidence that participants predicted these high-value social rewards. In Study 2, this finding held under experimental conditions that were free from the influence of extraneous variables, including conversation content and natural differences in smile predictability. Although results from naturalistic studies do not allow firm conclusions about the mechanisms governing smile reciprocity, they do hint that social-reward value may explain face-to-face behavior, given that the smiles people naturally see, compared with computerized laboratory stimuli, have clear and immediate social value.

Together with the naturalistic data, our EMG results suggest that different neural mechanisms control reciprocity, depending on the type of expression involved. Specifically, we argue that predictive reward-anticipation processes control genuine-smile reciprocity, whereas reactive mechanisms control polite-smile reciprocity. The social-reward properties of these stimuli likely cause this difference. Genuine smiles promote positive affect (Ekman et al., 1990) and predict future positive outcomes (Bernstein, Young, Brown, Sacco, & Claypool, 2008). Thus, genuine smiles carry intrinsic value as social reinforcers (Shore & Heerey, 2011) and lead to smile-specific anticipatory behavior. Their social-reward value is a likely explanation for why people learn faster in response to genuine smiles than polite smiles and show behavioral anticipation for genuine smiles only.

These data have important implications for understanding how the brain controls social interaction. Indeed, they imply that predictive or anticipatory processes play a fundamental role in the control of social behavior and suggest that response timing is important. For example, social cues that occur too quickly or too slowly may degrade social experience (Oberman, Winkielman, & Ramachandran, 2009). Thus, learning to anticipate genuine smiles may be an important social skill. Alterations in the timing of genuine-smile responses may affect social outcomes by violating expectations about when these rewards should appear. As reinforcement-learning research shows, violations of temporal expectations lead to changes in neuronal firing rates (Niv & Schoenbaum, 2008; Schultz, 2007) that are experienced as unpleasant (Jocham, Neumann, Klein, Danielmeier, & Ullsperger, 2009). Because healthy social partners find violations of social expectations unpleasant or confusing, such violations may damage social outcomes among, for example, people with neuropsychiatric conditions that interfere with reward-learning processes (Gradin et al., 2011).

Conclusions

We showed, under both experimental and observational conditions, that reactive and anticipatory mechanisms play important roles in driving social behavior. Our data

suggest that facial displays that carry social value are anticipated behaviorally, whereas displays that do not are mimicked only after they are perceptible. Thus, these results provide an important insight into the systems supporting smile reciprocity and represent a significant advance in understanding the mechanisms underpinning the neural control of real-world social behavior.

Author Contributions

E. A. Heerey developed the study concept, handled computer programming, and drafted the manuscript. H. M. Crossley completed data collection. Both authors designed the study, analyzed and interpreted the data, provided critical revisions, and approved the final version of the manuscript.

Declaration of Conflicting Interests

The authors declared that they had no conflicts of interest with respect to their authorship or the publication of this article.

Supplemental Material

Additional supporting information may be found at http://pss.sagepub.com/content/by/supplemental-data

Notes

1. An analysis using a more stringent time criterion (0–80 ms following the onset of a conversation partner's smile) yielded similar results: Genuine smiles were significantly more likely to be anticipated than were polite smiles, even when only these early time points were considered, paired-samples $t(69) = 6.94$, $p < .001$.

2. Although stimuli were carefully selected on the basis of discriminability statistics (see Shore & Heerey, 2011), this study used image sequences in which the faces morphed from neutral expressions to smiles. To ensure that smiles were not distinguishable on the basis of the amount of motion within those sequences, we used the method outlined in Schippers, Roebroeck, Renken, Nanetti, and Keysers (2010), to quantify movement across image frames. A paired-samples t test showed no evidence of movement differences across the genuine- and polite-smile sequences, $t(7) = -0.83$, $p = .44$.

References

Achaibou, A., Pourtois, G., Schwartz, S., & Vuilleumier, P. (2008). Simultaneous recording of EEG and facial muscle reactions during spontaneous emotional mimicry. *Neuropsychologia*, *46*, 1104–1113. doi:10.1016/j.neuropsychologia.2007.10.019

Averbeck, B. B., & Duchaine, B. (2009). Integration of social and utilitarian factors in decision making. *Emotion*, *9*, 599–608. doi:10.1037/a0016509

Bakeman, R., & Quera, V. (2011). *Sequential analysis and observational methods for the behavioral sciences*. Cambridge, England: Cambridge University Press.

Behrens, T. E., Hunt, L. T., & Rushworth, M. F. (2009). The computation of social behavior. *Science*, *324*, 1160–1164. doi:10.1126/science.1169694

Bernstein, M. J., Young, S. G., Brown, C. M., Sacco, D. F., & Claypool, H. M. (2008). Adaptive responses to social exclusion: Social rejection improves detection of real and fake smiles. *Psychological Science*, *19*, 981–983. doi:10.1111/j.1467-9280.2008.02187.x

Capella, J. (1997). Behavioral and judged coordination in adult informal social interactions: Vocal and kinesic indicators. *Journal of Personality and Social Psychology*, *72*, 119–131.

Chartrand, T. L., & Bargh, J. A. (1999). The chameleon effect: The perception-behavior link and social interaction. *Journal of Personality and Social Psychology*, *76*, 893–910.

Cialdini, R. B., & Goldstein, N. J. (2004). Social influence: Compliance and conformity. *Annual Review of Psychology*, *55*, 591–621.

Eder, D. (1993). "Go get ya a French!": Romantic and sexual teasing among adolescent girls. In D. Tannen (Ed.), *Gender and conversational interaction* (pp. 17–31). Oxford, England: Oxford University Press.

Ekman, P., Davidson, R. J., & Friesen, W. V. (1990). The Duchenne smile: Emotional expression and brain physiology: II. *Journal of Personality and Social Psychology*, *58*, 342–353.

Ekman, P., Sorenson, E. R., & Friesen, W. V. (1969). Pan-cultural elements in facial displays of emotion. *Science*, *164*, 86–88.

Fridlund, A. J. (1991). Sociality of solitary smiling: Potentiation by an implicit audience. *Journal of Personality and Social Psychology*, *60*, 229–240.

Fridlund, A. J., & Cacioppo, J. T. (1986). Guidelines for human electromyographic research. *Psychophysiology*, *23*, 567–589.

Goffman, E. (1955). On face-work: An analysis of ritual elements in social interaction. *Psychiatry*, *18*, 213–231.

Gradin, V. B., Kumar, P., Waiter, G., Ahearn, T., Stickle, C., Milders, M., . . . Steele, J. D. (2011). Expected value and prediction error abnormalities in depression and schizophrenia. *Brain*, *134*, 1751–1764. doi:10.1093/brain/awr059

Heerey, E. A., & Kring, A. M. (2007). Interpersonal consequences of social anxiety. *Journal of Abnormal Psychology*, *116*, 125–134. doi:10.1037/0021-843X.116.1.125

Hess, U., & Bourgeois, P. (2010). You smile—I smile: Emotion expression in social interaction. *Biological Psychology*, *84*, 514–520. doi:10.1016/j.biopsycho.2009.11.001

Heyes, C. (2011). Automatic imitation. *Psychological Bulletin*, *137*, 463–483. doi:10.1037/a0022288

Jocham, G., Neumann, J., Klein, T. A., Danielmeier, C., & Ullsperger, M. (2009). Adaptive coding of action values in the human rostral cingulate zone. *Journal of Neuroscience*, *29*, 7489–7496. doi:10.1523/JNEUROSCI.0349-09.2009

King-Casas, B., Tomlin, D., Anen, C., Camerer, C. F., Quartz, S. R., & Montague, P. R. (2005). Getting to know you: Reputation and trust in a two-person economic exchange. *Science*, *308*, 78–83. doi:10.1126/science.1108062

Kring, A. M., & Sloan, D. M. (2007). The Facial Expression Coding System (FACES): Development, validation, and utility. *Psychological Assessment*, *19*, 210–224. doi:10.1037/1040-3590.19.2.210

Krumhuber, E. G., & Manstead, A. S. (2009). Can Duchenne smiles be feigned? New evidence on felt and false smiles. *Emotion*, *9*, 807–820. doi:10.1037/a0017844

Mitchell, G. (2012). Revisiting truth or triviality: The external validity of research in the psychological laboratory. *Perspectives on Psychological Science*, *7*, 109–118.

Niv, Y., & Schoenbaum, G. (2008). Dialogues on prediction errors. *Trends in Cognitive Sciences*, *12*, 265–272. doi:10.1016/j.tics.2008.03.006

Oberman, L. M., Winkielman, P., & Ramachandran, V. S. (2009). Slow echo: Facial EMG evidence for the delay of spontaneous, but not voluntary, emotional mimicry in children with autism spectrum disorders. *Developmental Science*, *12*, 510–520. doi:10.1111/j.1467-7687.2008.00796.x

O'Doherty, J., Winston, J., Critchley, H., Perrett, D., Burt, D. M., & Dolan, R. J. (2003). Beauty in a smile: The role of medial orbitofrontal cortex in facial attractiveness. *Neuropsychologia*, *41*, 147–155.

Pickering, L., Corduas, M., Eisterhold, J., Seifried, B., Eggleston, A., & Attardo, S. (2009). Prosodic markers of saliency in humorous narratives. *Discourse Processes*, *46*, 517–540.

Repacholi, B., & Slaughter, V. (Eds.). (2003). *Individual differences in theory of mind: Implications for typical and atypical development*. New York, NY: Psychology Press.

Russell, J. A., Bachorowski, J. A., & Fernandez-Dols, J. M. (2003). Facial and vocal expressions of emotion. *Annual Review of Psychology*, *54*, 329–349. doi:10.1146/annurev.psych.54.101601.145102

Sanders, A. F. (1998). *Elements of human performance: Reaction processes and attention in human skill*. Mahwah, NJ: Erlbaum.

Scharlemann, J. P. W., Eckel, C. C., Kacelnik, A., & Wilson, R. K. (2001). The value of a smile: Game theory with a human face. *Journal of Economic Psychology*, *22*, 617–640.

Schippers, M., Roebroeck, A., Renken, R., Nanetti, L., & Keysers, C. (2010). Mapping the flow from one brain to another during gestural communication. *Proceedings of the National Academy of Sciences, USA*, *107*, 9388–9393.

Schultz, W. (2006). Behavioral theories and the neurophysiology of reward. *Annual Review of Psychology*, *57*, 87–115. doi:10.1146/annurev.psych.56.091103.070229

Schultz, W. (2007). Multiple dopamine functions at different time courses. *Annual Review of Neuroscience*, *30*, 259–288. doi:10.1146/annurev.neuro.28.061604.135722

Shore, D. M., & Heerey, E. A. (2011). The value of genuine and polite smiles. *Emotion*, *11*, 169–174. doi:10.1037/a0022601

Stel, M., & van Knippenberg, A. (2008). The role of facial mimicry in the recognition of affect. *Psychological Science*, *19*, 984–985. doi:10.1111/j.1467-9280.2008.02188.x

Stel, M., & Vonk, R. (2010). Mimicry in social interaction: Benefits for mimickers, mimickees, and their interaction. *British Journal of Psychology*, *101*, 311–323. doi:10.1348/000712609X465424

van Baaren, R. B., Holland, R. W., Kawakami, K., & van Knippenberg, A. (2004). Mimicry and prosocial behavior. *Psychological Science*, *15*, 71–74.

van Boxtel, A. (2003). Optimal signal bandwidth for the recording of surface EMG activity of facial, jaw, oral, and neck muscles. *Psychophysiology*, *38*, 22–34.

Watanabe, M. (1996). Reward expectancy in primate prefrontal neurons. *Nature*, *382*, 629–632.

Wild, B., Erb, M., Eyb, M., Bartels, M., & Grodd, W. (2003). Why are smiles contagious? An fMRI study of the interaction between perception of facial affect and facial movements. *Psychiatry Research*, *123*, 17–36.